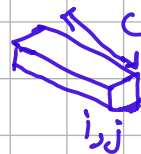


1)

$$a) \quad Y_{ij} = W X_{ij} \quad W \in \mathbb{R}^{C \times C}$$



$$\frac{\partial Y_{ij}}{\partial X_{kl}} = 0 \quad \text{if } k \neq i \text{ or } j \neq l$$

$$\text{vec}(X) = (X_{000}, X_{100}, \dots, X_{C,00}, \dots, X_{0HW}, \dots, X_{CHW})^T$$

$$J_g(x) \in \mathbb{R}^{CHW \times CHW}$$

$$J_g(x)_{c'ij, cij} = W_{cc'} X_{c'ij} \delta(c, c') \delta(i, i') \delta(j, j')$$

$$\Rightarrow J_g(x)_{ij, ij} = W \delta(i, i') \delta(j, j') \Rightarrow J_g(x) = \begin{bmatrix} W & 0 & \dots & 0 \\ 0 & W & & \\ \vdots & & \ddots & \\ 0 & & & W \end{bmatrix}$$

$$\Rightarrow \det J_g(x) = \prod_{i=1}^{HW} \det W = \det W^{HW}$$

$$g(x) \text{ is invertible} \Leftrightarrow \det J_g(x) \neq 0 \Leftrightarrow \det W \neq 0$$

$$\det W \neq 0 \Rightarrow W \text{ is invertible: } W^{-1}$$

$$Y_{ij} = W X_{ij} \Rightarrow X_{ij} = W^{-1} Y_{ij} \quad \square$$

$$b) \quad W = PL(U + \text{diag}(S))$$

$$\det W = \det P \det L \det(U + \text{diag}(S))$$

$$\det L = 1 \quad \det(U + \text{diag}(S)) = \prod_{i=1}^C s_i$$

$$\det P = (-1)^m \Rightarrow |\det W| = \left| \prod_{i=1}^C s_i \right| = \prod_{i=1}^C |s_i|$$

$$\Rightarrow \log |\det W| = \sum_{i=1}^c |s_i|$$

c)

$$Wz = b \quad W = LU$$

$$e_i = (0 \dots 1 \dots 0)^T$$

$$Wz = e_i \Rightarrow LUz = e_i$$

$$Uz = y \Rightarrow Ly = e_i$$

$$Ly = \begin{bmatrix} \triangle \\ \vdots \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_c \end{bmatrix}$$

y can be calculated using forward substitution in $\mathcal{O}(n^2)$ ops.

$$Uz = y \quad Uz = \begin{bmatrix} \triangle \\ \vdots \end{bmatrix} \begin{bmatrix} z_1 \\ \vdots \\ z_c \end{bmatrix}$$

We have y .

z can be calculated using backward substitution in $\mathcal{O}(n^2)$ ops.

$$Wz = e_i$$

$$WW^{-1} = \begin{bmatrix} WW_1^{-1} & WW_2^{-1} & \dots & WW_c^{-1} \end{bmatrix} = \begin{bmatrix} e_1 & \dots & e_c \end{bmatrix}$$

$$\Rightarrow z = W^{-1} \quad Wz_i = e_i \Rightarrow W^{-1} = \begin{bmatrix} z_1 & \dots & z_c \end{bmatrix}$$

$\Rightarrow$ we can calculate each z_i in $\mathcal{O}(c^2)$

$\Rightarrow$ we can calculate W^{-1} in $\mathcal{O}(c^3)$.

2)

a.i) $z_0 \sim p(z_0)$

$$\frac{dz}{dt} = f_\theta(z_{t+1}, t)$$

$$z(t) = f(z(t')) \Rightarrow p(z(t)) = p(z(t')) | \det J_f(z(t')) |^{-1}$$

$$\Rightarrow \log p(z(t)) = \log p(z(t')) - \log | \det J_f(z(t')) |$$

$$= \log p(g(z(t))) - \log | \det J_f(g(z(t))) |$$

$$z(t') = g(z(t))$$

a.ii)

$$\frac{d}{dt} \log p(z(t)) = \frac{1}{p(g(z(t)))} g'(z(t)) \dot{z}(t) - \frac{d}{dt} \log | \det J_f(g(z(t))) |$$

a.iii)

$$\frac{d}{dt} \log | \det J_f(g(z(t))) | = \text{Tr} \left(J_f^{-1}(t) \frac{d}{dt} J_f(t) \right)$$

$$\Rightarrow \frac{d}{dt} \log p(z(t)) = \frac{g'(z(t))}{p(g(z(t)))} f_\theta(z(t), t) -$$

$$\text{Tr} \left(J_f^{-1}(t) \frac{d}{dt} J_f(t) \right)$$

$$\frac{d}{dt} \log |x(t)| = \frac{1}{|x(t)|} \dot{x}(t) \text{sign}(x(t)) = \frac{\dot{x}(t)}{x(t)}$$

$$A = Q \Lambda Q^T$$

$$\dot{A} = \dot{Q} \Lambda Q^T + Q \dot{\Lambda} Q^T + Q \Lambda \dot{Q}^T$$

$$\Rightarrow A^T \dot{A} = Q \Lambda^{-1} \dot{Q}^T (\dot{Q} \Lambda Q^T + Q \dot{\Lambda} Q^T + Q \Lambda \dot{Q}^T)$$

$$= Q \Lambda^{-1} \dot{Q}^T Q \Lambda Q^T + Q \Lambda^{-1} \dot{\Lambda} Q^T + Q \dot{Q}^T$$

$$\Rightarrow \text{tr}(A^T \dot{A}) = \text{tr}(Q \Lambda^{-1} \dot{Q}^T Q \Lambda Q^T) + \text{tr}(Q \Lambda^{-1} \dot{\Lambda} Q^T) + \text{tr}(Q \dot{Q}^T)$$

$$= \text{tr}(\underbrace{\dot{Q} Q^T + Q \dot{Q}^T}_{\frac{d}{dt}(Q Q^T) = 0}) + \sum_i \frac{\dot{\lambda}_i}{\lambda_i} = \sum_i \frac{1}{\lambda_i} \frac{d \lambda_i}{dt}$$

$$\frac{d}{dt}(Q Q^T) = 0$$

$$= \sum_i \frac{d}{dt} \log |\lambda_i|$$

$$= \frac{d}{dt} \log \prod_i |\lambda_i|$$

$$= \frac{d}{dt} \log |\det A(t)|$$

□

$$\text{a.ii)} \quad \frac{d}{dt} J_{f(t)}{}_{ij} = \frac{d}{dt} \frac{df(t)_i}{dg(t)_j} = \frac{d}{dt} \frac{dz(t)_i}{dz(t)_j} = \frac{dz(t)_i}{dz(t)_j}$$

$$= \frac{df_0(z(t), t)_i}{dg(t)_j} = \sum_k \frac{df_0_i}{dz(t)_k} \frac{dz(t)_k}{dz(t)_j}$$

$$= \left(\frac{\partial f_0}{\partial z} \right)_{i,:} J_{f(t)}{}_{:,j}$$

$$\frac{\partial f_0}{\partial z} = A$$

$$J_{f(t)} = B$$

$$\Rightarrow \frac{d}{dt} J_{f(t)} = AB = \frac{\partial f_0}{\partial z} J_{f(t)}$$

□

$$\text{a.v)} \quad \text{Tr} \left(J_{f(t)}^{-1} \frac{d}{dt} J_{f(t)} \right) = \text{Tr} \left(J_{f(t)}^{-1} \frac{\partial f_0}{\partial z} J_{f(t)} \right)$$

$$= \text{Tr} \left(J_{f(t)} J_{f(t)}^{-1} \frac{\partial f_0}{\partial z} \right) = \text{Tr} \left(\frac{\partial f_0}{\partial z} \right)$$

$$\Rightarrow \frac{d}{dt} \log p(z(t)) = \frac{g'(z(t))}{p(g(z(t)))} f_\theta(z(t), t) -$$

$$\text{Tr} \left(J_{f(t)}^{-1} \frac{d}{dt} J_{f(t)} \right)$$

$$t'=0 \Rightarrow \frac{g'(z(t))}{p(g(z(t)))} f_\theta(z(t), t) = \frac{d}{dt} (\log p(g(z(t))))$$

$$p(z(0))=1$$

$$= \frac{d}{dt} (\log p(z(0))) = \frac{d}{dt} (0) = 0$$

$$\Rightarrow \frac{d}{dt} \log p(z(t)) = - \text{Tr} \left(J_{f(t)}^{-1} \frac{d}{dt} J_{f(t)} \right) = - \text{Tr} \left(\frac{\partial f_\theta}{\partial z} \right) \quad \square$$

3)

$$a) \quad x_i = z_i \times \exp(\alpha(x_{<i})) + \mu(x_{<i}) \quad \text{forward transformation}$$

$$z_i = \frac{x_i - \mu(x_{<i})}{\exp(\alpha(z_{<i}))} \quad \text{Backward transformation}$$

$$J_f(z)_{ij} = \frac{\partial f(z)_i}{\partial z_j} = \frac{\partial x_i}{\partial z_j} = \begin{cases} 0 & j < i \\ \exp(\alpha(x_{<i})) & i=j \\ \frac{\partial x_i}{\partial z_j} & \text{o.w.} \end{cases}$$

$$\Rightarrow J_f(z) = \begin{bmatrix} \exp(\alpha(x_{<1})) & & 0 \\ \vdots & \ddots & \\ \frac{\partial x_D}{\partial z_1} & \dots & \exp(\alpha(x_{<D})) \end{bmatrix} \Rightarrow \det J_f(z) = \prod_{i=1}^D \exp(\alpha(x_{<i}))$$

$$\Rightarrow \log |\det J_f(z)| = \sum_{i=1}^D \alpha(x_{<i})$$

$$p(x) = p(z) |\det J_f(z)|^{-1} = \frac{p(z)}{\prod_{i=1}^D \exp(\alpha(x_{<i}))}$$

$$\begin{aligned} \Rightarrow \log p(x) &= \log p(z) - \sum_{i=1}^D \alpha(x_{<i}) \quad z \sim \mathcal{N}(0, I) \\ &= -\frac{D}{2} \log(2\pi) - \frac{1}{2} \|z\|^2 - \sum_{i=1}^D \alpha(x_{<i}) \end{aligned}$$

$$x = f(z) = f_k(z^k) = f_k \circ f_{k-1} \circ \dots \circ f_1$$

$$\Rightarrow \log p_{\text{MAF}}(x) = \log p(\underbrace{z^k}_x) = \log p(z^{k-1}) - \sum_{i=1}^D \alpha_k(z^k_{<i})$$

$$\Rightarrow \log p_{\text{MAF}}(x) = \log p(z^0) - \sum_{k=1}^K \sum_{i=1}^D \alpha_k(z_{<i}^k) \quad \begin{matrix} z^{(0)} \sim \mathcal{N}(0, I) \\ z^{(K)} = x \end{matrix}$$

$$= -\frac{D}{2} \log(2\pi) - \frac{1}{2} \|z^{(0)}\|^2 - \sum_{k=1}^K \sum_{i=1}^D \alpha_k(z_{<i}^k)$$

Likelihood: with a single forward pass given $x \in \mathbb{R}^D$ we have

all $\alpha_k(z_{<i}^k)$ and $\mu_k(z_{<i}^k)$ are calculated with $z_i^{k-1} = \frac{z_i^k - \mu_k(z_{<i}^k)}{\exp(\alpha_k(z_{<i}^k))}$

$\Rightarrow \log p_{\text{MAF}}(x)$ can be calculated in parallel.

Sampling: 1. $z^{(0)} \sim \mathcal{N}(0, I)$

$$2. z_i^k = z_i^{k-1} \exp(\alpha_k(z_{<i}^k)) + \mu_k(z_{<i}^k)$$

$\Rightarrow$ we need to calculate each z_i^k sequentially.

$$b) x_i = z_i \exp(\alpha(z_{<i})) + \mu(z_{<i}) \quad \text{forward transformation}$$

$$z_i = \frac{x_i - \mu(z_{<i})}{\exp(\alpha(z_{<i}))}$$

Backward transformation

$$J_f(z)_{ij} = \frac{\partial f(z)_i}{\partial z_j} = \frac{\partial x_i}{\partial z_j} = \begin{cases} 0 & j < i \\ \exp(\alpha(z_{<i})) & i=j \\ \frac{\partial x_i}{\partial z_j} & \text{o.w.} \end{cases}$$

$$\Rightarrow J_f(z) = \begin{bmatrix} \exp(\alpha(z_{<1})) & & 0 \\ \vdots & \ddots & \\ \frac{\partial x_0}{\partial z_1} & \dots & \exp(\alpha(z_{<0})) \end{bmatrix} \Rightarrow \det J_f(z) = \prod_{i=1}^D \exp(\alpha(z_{<i}))$$

$$\Rightarrow \log |\det J_f(z)| = \sum_{i=1}^D \psi(z_{<i})$$

$$P(x) = p(z) |\det J_f(z)|^{-1} = \frac{p(z)}{\prod_{i=1}^D \exp(\alpha(z_{<i}))}$$

$$\begin{aligned} \Rightarrow \log P(x) &= \log p(z) - \sum_{i=1}^D \alpha(z_{<i}) \quad z \sim \mathcal{N}(0, I) \\ &= -\frac{D}{2} \log(2\pi) - \frac{1}{2} \|z\|^2 - \sum_{i=1}^D \alpha(z_{<i}) \end{aligned}$$

In IAF x_i is conditioned on previous latent vectors instead of generated $x_{<i}$.

Likelihood: given an input $x \in \mathbb{R}^D$ we don't have z_i^k

and by inverse transformation we have:

$$z_i^k = \frac{x_i - \mu_k(z_{<i}^k)}{\exp(\alpha_k(z_{<i}^k))} \Rightarrow \text{we need } \mu_k(z_{<i}^k) \text{ and } \alpha_k(z_{<i}^k) \text{ before we have } z_i^k.$$

$\Rightarrow$ Likelihood is estimated sequential.

sampling: 1. $z^{(0)} \sim \mathcal{N}(0, I)$

$$2. z_i^k = z_i^{k-1} \exp(\alpha_k(z_i^{k-1})) + \mu_k(z_i^{k-1})$$

all z_i^{k-1} are available $\Rightarrow z_i^k$ can be calculated in parallel.

$$c) D_{KL}(\pi_x(x) \parallel p_{MAF}(x)) = \mathbb{E}_{x \sim \pi_x} \left[\log \left(\frac{\pi_x(x)}{p_{MAF}(x)} \right) \right]$$

$$= \int \log \left(\frac{\pi_x(x)}{p_{MAF}(x)} \right) \pi_x(x) dx$$

$$\pi_x(x) = p_u(u) |\det J_f(u)|^{-1}$$

$$x = f(u) \Rightarrow dx = |\det J_f(u)| du$$

$$p_{\text{MAF}}(x) = \pi_u(u) |\det J_f(u)|^{-1}$$

$$\Rightarrow D_{KL}(\pi_x(u) \| p_{\text{MAF}}(u)) = \int_u \log\left(\frac{p_u(u)}{\pi_u(u)}\right) p_u(u) |\det J_f(u)|^{-1} |\det J_f(u)| du$$

$$= \mathbb{E}_{u \sim p_u} \left[\log \frac{p_u(u)}{\pi_u(u)} \right] = D_{KL}(p_u(u) \| \pi_u(u))$$

□

4)

ai, aii)

$$L = \iint \tilde{p}(x) \int \delta(z - F_x(u)) \mathcal{N}(u; 0, I) du \log \frac{\tilde{p}(x) \int \delta(z - F_x(u)) \mathcal{N}(u; 0, I) du}{q(x|z) q(z)} dz dx$$

$$= \int \tilde{p}(x) \iint \delta(z - F_x(u)) \mathcal{N}(u; 0, I) \log \frac{\tilde{p}(x) \int \delta(z - F_x(u)) \mathcal{N}(u; 0, I) du}{q(x|z) q(z)} dz du dx$$

$$= \int \tilde{p}(x) \int \mathcal{N}(u; 0, I) \log \frac{\tilde{p}(x) \int \delta(F_x(u) - F_x(u')) \mathcal{N}(u'; 0, I) du'}{q(x|z) q(z)} du dx$$

$$\int \delta(F_x(u) - \underbrace{F_x(u')}_{v}) \mathcal{N}(u'; 0, I) du' = \int \delta(F_x(u) - v) \mathcal{N}(H_x(v); 0, I) \left| \det \frac{\partial F_x(u')}{\partial u'} \right|^{-1} dv$$

$$dv = du' \left| \det \frac{\partial F_x(u')}{\partial u'} \right|$$

$$= \mathcal{N}(\underbrace{H_x(F_x(u))}_u; 0, I) \left| \det \frac{\partial F_x(u')}{\partial u'} \right|^{-1} \Big|_{u'=u}$$

$$\Rightarrow L = \iint \tilde{p}(x) \mathcal{N}(u; 0, I) \log \left(\frac{\tilde{p}(x) \mathcal{N}(u; 0, I)}{q(x|F_x(u)) q(F_x(u))} \left| \det \frac{\partial F_x(u')}{\partial u'} \right|^{-1} \Big|_{u'=u} \right) du dx$$

$\rightarrow \mathcal{N}(0, I) = q(z)$

aiii)

$$\bullet v = F_x(u) \Rightarrow dv = du \left| \det \frac{\partial F_x(u)}{\partial u} \right|$$

The term $\left| \det \frac{\partial F_x(u)}{\partial u} \right|^{-1}$ comes from the transformation

from $u \rightarrow v$.

we have to scale v according to $J_{F_x}(u)$.

• By simply setting $F_x(u) = s * u$ we have:

$$p(z|x) = \int \delta(z - s * u) \mathcal{N}(u; 0, I) du = \int \delta(z - v) \mathcal{N}(s^{-1} * v; 0, I) s^{-1} * dv$$

$$\Rightarrow p(z|x) = s^{-1} * \mathcal{N}(s^{-1}z; 0, I) \quad \text{we can have any } \mathcal{N}(0, \text{diag}(s^{-1})).$$

b i)

$$q(x|F_x(u)) = \mathcal{N}(x; F_x^{-1}(F_x(u)), \sigma^2 I) \\ = \mathcal{N}(x; u, \sigma^2 I)$$

$$\Rightarrow \log q(x|F_x(u)) = -\frac{d}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|u\|_2^2$$

$$\text{b ii)} \quad J_{F_x}(u)_{ij} = \frac{\partial F_x(u)_i}{\partial u_j} = \frac{\partial F(\sigma u + x)_i}{\partial u_j} = \sum_k \frac{\partial F(u)_i}{\partial w_k} \frac{\partial w_k}{\partial u_j}$$

$$w = \sigma u + x \Rightarrow \frac{\partial w_k}{\partial u_j} = \sigma \delta_{kj} \Rightarrow J_{F_x}(u)_{ij} = \frac{\partial F(u)_i}{\partial w_j} \sigma$$

$$\Rightarrow \log \left| \det \frac{\partial F_x(u)}{\partial u} \right| = \log \left| \det \left(\sigma \frac{\partial F(u)}{\partial w} \right) \right| = \log \left| \sigma^d \det \frac{\partial F(u)}{\partial w} \right| \\ = d \log \sigma + \log \left| \det \frac{\partial F(u)}{\partial w} \right|$$

b iii)

$$\mathcal{L} = \int \tilde{p}(x) \log \tilde{p}(x) dx + \int \mathcal{N}(u; 0, I) \log(\mathcal{N}(u; 0, I)) du$$

$$= \mathbb{E}_{\tilde{p}(x), u \sim \mathcal{N}(0, I)} \left[\log q(x|F_x(u)) + \log \mathcal{N}(F_x(u); 0, I) + \log \left| \det \frac{\partial F_x(u)}{\partial u} \right|^{-1} \right] \\ = \mathbb{E}_{\tilde{p}(x), u \sim \mathcal{N}(0, I)} \left[\underbrace{\frac{d}{2} \log(2\pi\sigma^2)}_{\text{Const}} + \underbrace{\frac{1}{2\sigma^2} \|u\|_2^2}_{\text{Const}} + \underbrace{\frac{d}{2} \log(2\pi)}_{\text{Const}} + \underbrace{\frac{1}{2} \|F(\sigma u + x)\|_2^2}_{\text{Const}} \right. \\ \left. - d \log \sigma - \log \left| \det \frac{\partial F(u)}{\partial w} \right| \right] + \text{Const}$$

$$\Rightarrow \mathcal{L} = \mathbb{E}_{\tilde{p}(x), u \sim \mathcal{N}(0, I)} \left[\frac{1}{2} \|F(\sigma u + x)\|_2^2 - \log \left| \det \frac{\partial F(u)}{\partial w} \right| \right] + \text{Const}$$

5)

$$a) A = \emptyset \Rightarrow \bar{A} = B \Rightarrow P(\bar{A}) = P(B) = 0.9 \times 1$$

$$\Rightarrow \forall x, y \in \mathbb{R} \quad P_\theta(x, y) < 0.9 - \epsilon \Rightarrow P_\theta(\bar{A}) = P_\theta(B) < (0.9 - \epsilon) \times 1$$

$$\Rightarrow |P_\theta(\bar{A}) - P(\bar{A})| > \epsilon$$

$$\text{Pinker's inequality: } \sqrt{\frac{1}{2} D_{KL}(P_\theta \| P)} \geq \sup_E \{ |P_\theta(E) - P(E)| \} \\ \geq P_\theta(\bar{A}) - P(\bar{A}) > \epsilon$$

$$\Rightarrow D_{KL}(P_\theta \| P) > 2\epsilon^2$$

$$b) C = \{z \in \mathbb{R}^2; f_\theta^{-1}(z) \in A\} \quad dx = dz \left| \det \frac{\partial f_\theta(z)}{\partial z} \right|^{-1}$$

$$\text{Vol}(A) = \int_A dx = \int_C \underbrace{\left| \det \frac{\partial f_\theta(z)}{\partial z} \right|^{-1}}_{=1} dz = \int_C dz = \text{Vol}(C)$$

$$P_\theta(x) = P(z) \Rightarrow \forall z \in C \quad P(z) \geq 0.9 - \epsilon \quad \epsilon > 0$$

$$z \sim \mathcal{N}(0, I) \Rightarrow P(z) = \frac{1}{(\sqrt{2\pi})^2} \exp\left(-\frac{\|z\|_2^2}{2}\right) \geq 0.9 - \epsilon \quad \forall z \in C$$

$$\left. \begin{aligned} P_\theta(A) &\geq (0.9 - \epsilon) \text{Vol}(A) \\ P(C) &\leq \frac{1}{2\pi} \pi \left(-2 \log(2\pi(0.9 - \epsilon)) \right) \end{aligned} \right\} \Rightarrow \text{Vol}(A) \leq -\log(2\pi(0.9 - \epsilon))$$

$$c) P(B) = 0.9 \times 1$$

$$P_\theta(B) = P(B) = \int_B P(z) dz \leq \frac{1}{2\pi} \int_B dz \stackrel{\rightarrow 1}{=} \frac{\text{Vol}(B)}{2\pi} = \frac{1}{2\pi}$$

$$\Rightarrow |P_\theta(B) - P(B)| = |0.9 - \frac{1}{2\pi}| > 0 \Rightarrow D_{KL}(P_\theta \| P) \geq 2(0.9 - \frac{1}{2\pi})^2 > 0$$

d)

$$D_{KL}(P_\theta \| P) \geq 2(0.9 - \frac{1}{2\pi})^2 \Rightarrow \lim_{n \rightarrow \infty} P_{\theta_n}(x) - P(x) \neq 0$$

$\Rightarrow P_{\theta_n}(x)$ can never be a universal approximator.

